MONASAN INJECTION 1ML AMPOULE

DESCRIPTION: A pale yellow oily liquid.

COMPOSITION:
Each ampoule contains Fluphenazine Decanoate 25 mg per ml in sesame oil.

PHARMACODYNAMICS:
The basic effects of fluphenazine decanoate appear to be no different from those of fluphenazine hydrochloride, with the exception of duration of action. The esterification of fluphenazine markedly prolongs the drug's duration of effect without unduly attenuating its beneficial action. Fluphenazine has activity at all levels of the central nervous system as well as on multiple organ systems. The mechanism whereby its therapeutic action is exerted is unknown. Fluphenazine differs from other phenothiazine-derivatives in several respects: it is more potent on a milligram basis, it has less potentiating effect on central nervous system depressants and anaesthetics than do some of the phenothiazines and appears to be less sedating, and it is less likely than some of the older phenothiazines to produce hypotension (nevertheless appropriate cautions should be observed - see sections on *Precautions and Adverse Effects*).
Fluphenazine is a piperazine phenothiazine used widely in the treatment of schizophrenia. The therapeutic action of fluphenazine is thought to be through dopamine receptor blockade in the mesolimbic pathways. It has weak anticholinergic and sedative effects, antiadrenergic and neuroendocrine actions of strong extrapyramidal effects.

PHARMACOKINETICS:
When Fluphenazine is esterified to form fluphenazine decanoate, formulated in an oil vehicle and injected deeply into muscle, the ester slowly diffuses into the circulation. After hydrolysis the free form of the drug is released which can cross the blood-brain barrier.
Following intramuscular injection of fluphenazine decanoate in sesame oil the onset of action occurs within 24-72 hours; the duration of action is usually 1-6 weeks, with an average of 2 weeks. The distribution and metabolic rate of fluphenazine have not been fully elucidated. In one study, the plasma elimination half-life of fluphenazine was 7-10 days, after intramuscular fluphenazine decanoate. Excretion of unchanged drug and metabolites occurred in urine and faeces.

INDICATION: For treatment and maintenance of schizophrenia

RECOMMENDED DOSAGE:
Fluphenazine decanoate injection is given intramuscularly. A dry syringe and needle of at least 21 gauge should be used. Use of a wet needle or syringe may cause the solution to become cloudy. To begin therapy with fluphenazine decanoate injection the following regimens are suggested.
For most patients, a dose of 12.5 to 25 mg (0.5 to 1 mL) may be given to initiate therapy. The onset of action generally appears between 24 and 72 hours after injection and the effects of the drug on psychotic symptoms become significant within 48 to 96 hours. Subsequent injections and the dosage interval are determined in accordance with the patient's response. When administered as maintenance therapy, a single injection may be effective in controlling schizophrenic symptoms up to four weeks or longer. The response to a single dose has been found to last as long as six weeks in a few patients on maintenance therapy. It may be advisable that patients who have had no history of taking phenothiazines should be treated initially with a shorter acting form of fluphenazine before administering the decanoate to determine the patient's response to fluphenazine and to establish appropriate dosage. Since the dosage comparability of the shorter acting forms of fluphenazine to the longer-active decanoate is not known, special caution should be exercised when switching from the shorter-acting forms to the decanoate.
Severely agitated patients may be treated with a rapid-acting phenothiazine compound. When acute symptoms have subsided, 25 mg (1 mL) of fluphenazine decanoate injection may be administered. Subsequent dosage is adjusted as necessary.
'Poor risk' patients (those with known hypersensitivity to phenothiazines or with disorders that predispose to undue reactions): Therapy may be initiated cautiously with oral or parenteral fluphenazine hydrochloride. When the pharmacologic effects and an appropriate dosage are apparent, an equivalent dose of fluphenazine decanoate injections may be administered. Subsequent dosage adjustments are made in accordance with the response of the patient. The optimal amount of the drug and the frequency of administration must be determined for each patient, since dosage requirements have been found to vary with clinical circumstances a well as with individual response to the drug.
Geriatric Patients: Doses in the lower range (1/4 to 1/3 of those in younger adults) may be sufficient for most elderly patients.
Dosage should not exceed 100 mg. If doses greater than 50 mg are deemed necessary, the next dose and succeeding doses should be increased cautiously in increments of 12.5 mg.

As this preparation contains benzyl alcohol, its use should be avoided in children under two years of age. Not to be used in neonates.

ROUTE OF ADMINISTRATION: For i.m. injection.

CONTRAINDICATIONS:
Phenothiazines are contraindicated in patients with suspected or established sub-cortical brain damage. Phenothiazine compounds should not be used in patients receiving large doses of hypnotics. Fluphenazine is contraindicated in comatose or severely depressed states. The presence of blood dyscrasia, liver damage or renal insufficiency precludes the use of fluphenazine decanoate. Fluphenazine decanoate is not intended for use in children under 12 years of age.
Fluphenazine decanoate injection is contraindicated in patients who have shown hypersensitivity to fluphenazine: cross sensitivity to phenothiazine derivates may occur.

WARNINGS & PRECAUTIONS:
Warnings. The use of this drug may impair the mental and physical abilities required for driving a car or operating heavy machinery. Physicians should be alert to the possibility that severe adverse reactions may occur which require immediate medical attention. Potentiation of the effects of alcohol may occur with the use of this drug. Since there is no adequate experience in children who have received this drug, safety and efficacy in children have not been established.
Precautions: Because of the possibility of cross sensitivity, fluphenazine decanoate should be used cautiously in patients who have developed jaundice, dermatoses, or other allergic reactions to phenothiazine derivatives. Psychotic patients on large doses of a phenothiazine drug who are undergoing surgery should be watched carefully for possible hypotensive phenomena. Moreover, it should be remembered that reduced amounts of anaesthetics or central nervous system depressants may be necessary.
Fluphenazine decanoate should be used cautiously in patients exposed to extreme heat or phosphorus insecticides.
The preparation should be used with caution in patients with a history of convulsive disorders, since grand mal convulsions have been known to occur. Use with caution in patients with special medical disorders such as mitral insufficiency or other cardiovascular diseases and pheochromocytoma. The possibility of liver damage, pigmentary retinopathy, lenticular and corneal deposits and development of irreversible dyskinesia should be remembered when patients are on prolonged therapy.
Outside state hospitals, or other psychiatric institutions, fluphenazine decanoate should be administered under the direction of a physician experienced in the clinical use of psychotropic drugs, particularly phenothiazine derivatives. Furthermore, facilities should be available for periodic checking of hepatic function, renal function and the blood picture. Renal function of patients on long-term therapy should be monitored; if BUN (blood urea nitrogen) becomes abnormal, treatment should be discontinued. As with any phenothiazine the physician should be alert to the possible development of 'silent pneumonias' in patients under treatment with fluphenazine decanoate.

INTERACTION WITH OTHER MEDICAMENTS:
1. CNS depression producing medications: Concurrent use with phenothiazines may result in increased CNS and respiratory depression and increased hypotensive effects.
2. Alcohol may increase the risk of heat stroke when taken concurrently with phenothiazines.
3. Concurrent use with tricyclic antidepressants or Maprotiline or monoamine oxidase (MAO) inhibitors including furazolidone, procarbazine and selegiline may prolong and intensify the sedative and anti-cholinergic effects of either these medications or phenothiazines.
   Phenothiazines may increase the plasma concentrations of cyclic depressants by inhibiting metabolism, conversely cyclic antidepressants may inhibit phenothiazine metabolism; also the risk of neuroleptic malignant syndrome (NMS) may be increased.
4. The concurrent use of antithyroid agents with phenothiazines may increase the risk of agranulocytosis.
5. Use of epinephrine to treat phenothiazine-induced hypotension may cause severe hypotension and tachycardia because the alpha-adrenergic effects of epinephrine may be blocked leaving only the beta stimulation.
6. Extrapyramidal reaction-causing medications: Concurrent use with phenothiazines may increase the severity and frequency of extrapyramidal effects.
7. Hypotension-producing medications: Use with phenothiazines may produce severe hypotension with postural syncope.
8. Antiparkinsonian effects of Levodopa may be inhibited when used with phenothiazines because of blockade of dopamine receptors in the brain and has not been shown to be effective in the treatment of phenothiazine-induced parkinsonism.

USE IN PREGNANCY & LACTATION:
Use in Pregnancy: The safety for the use of this drug during pregnancy has not been established; therefore the possible hazards should be weighed against the potential benefits when administering the drug to pregnant patients. When given in high doses during late pregnancy, phenothiazines have caused prolonged extrapyramidal disturbances in the child.
Neonates exposed to antipsychotic drugs during the third trimester of pregnancy are at risk for extrapyramidal and/or withdrawal symptoms following delivery. There have been reports of agitation, hypertonia, hypotonia, tremor, somnolence, respiratory distress, and feeding disorder in these neonates. These complications have varied in severity; while in some cases symptoms have been self-limited, in other cases neonates have required intensive care unit support and prolonged hospitalization.
Monasan Injection (1ml amp) should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

SIDE EFFECTS:

Central Nervous System: The side effects most frequently reported with phenothiazine compounds are extrapyramidal symptoms including pseudo-parkinsonism, dystonia, dyskinesia, akathisia, oculogyric crises, opisthotonos, and hyperreflexia. Most often these extrapyramidal symptoms are reversible; however, they may be persistent (see below). The frequency of such reactions is related in part to chemical structure; one can expect a higher incidence with fluphenazine decanoate than with less potent piperazine derivatives or with straight-chain phenothiazines such as chlorpromazine. With any given phenothiazine derivative the incidence and severity of such reactions depend more on individual patient sensitivity than on other factors, but dosage level and patient age are also determinants.

Extrapyramidal reactions may be alarming, and the patient should be forewarned and reassured. These reactions can usually be controlled by administration of antiparkinsonian drugs, such as Benztropine Mesylate, and by subsequent reduction in dosage.

Persistent tardive dyskinesia: As with all antipsychotic agents, tardive dyskinesia may appear in some patients on long-term therapy and may occur after drug therapy has been discontinued. The risk seems to be greater in elderly patients on high-dose therapy, especially females. The symptoms are persistent and in some patients appear to be irreversible. The syndrome is characterised by rhythmical involuntary movements of the tongue, face, mouth or jaw (e.g. protrusion of tongue, puffing of cheeks, puckering of mouth, chewing movements). Sometimes, these may be accompanied by involuntary movements of the extremities. There is no known effective treatment for tardive dyskinesia; antiparkinsonism agents usually do not alleviate the symptoms of this syndrome. It is suggested that all antipsychotic agents be discontinued if these symptoms appear. Should it be necessary to reinstitute treatment, or increase the dosage of the agent, or switch to a different antipsychotic agent, the syndrome may be masked. It has been reported that fine vermicular movements of the tongue may be an early sign of the syndrome and if the medication is stopped at that time, the syndrome may not develop.

Drowsiness or lethargy, if they occur, may necessitate a reduction in dosage; the induction of a catatonic-like state has been known to occur with dosages of fluphenazine far in excess of the recommended amounts. As with other phenothiazine compounds, reactivation or aggravation of psychotic processes may be encountered. Phenothiazine derivatives have been known to cause, in some patients, restlessness, excitement or bizarre dreams.

Autonomic Nervous System: Hypertension and fluctuations in blood pressure have been reported with fluphenazine. Hypotension has rarely presented a problem with fluphenazine. However, patients with pheochromocytoma cerebral vascular or renal insufficiency, or a severe cardiac reserve deficiency such as mitral insufficiency appear to be particularly prone to hypotensive reactions with phenothiazine compounds, and should therefore be observed closely when the drug is administered. If severe hypotension should occur, supportive measures including the use of intravenous vasopressor drugs should be instituted immediately. Adrenaline should not be used since phenothiazine derivatives have been found to reverse its action, resulting in a further lowering of blood pressure.

Autonomic reactions including nausea and loss of appetite, salivation, polyuria, perspiration, dry mouth, headache and constipation may occur. Autonomic effects can usually be controlled by reducing or temporarily discontinuing dosage. In some patients, phenothiazine derivatives have caused blurred vision, glaucoma, bladder paralysis, faecal impaction, paralytic ileus, tachycardia, or nasal congestion.

Metabolic and Endocrine. Weigh change, peripheral oedema, abnormal lactation, gynaecomastia, menstrual irregularities, false results on pregnancy tests, impotency in men and increased libido in women have all been known to occur in some patients on phenothiazine therapy.

Allergic Reactions. Skin disorders such as itching, erythema, urticaria, seborrhoea, photosensitivity, eczema and even exfoliative dermatitis have been reported with phenothiazine derivatives. The possibility of anaphylactoid reactions occurring in some patients should be borne in mind.

Haemotologic. Routine blood counts are advisable during therapy since blood dyscrasias including leukopenia, agranulocytosis, thrombocytopenic or non-thrombocytopenic purpura, eosinophilia and pancytopenia have been observed with phenothiazine derivatives. Furthermore, if any soreness of the mouth, gums or throat, or any symptoms of upper respiratory infection occur and confirmatory leukocyte count indicates cellular depression, therapy should be discontinued and other appropriate measured instituted immediately.

Hepatic. Liver damage manifested by cholestatic jaundice may be encountered, particularly during the first month of therapy; treatment should be discontinued if this occurs. An increase in cephalin flocculation, sometimes accompanied by alterations in other liver function tests, have been reported in patients receiving the enanthate ester of fluphenazine (a closely related compound) who have had no clinical evidence of liver damage).

Others. Sudden, unexpected and unexplained deaths have been reported in hospitalised psychotic patients receiving phenothiazines. Previous brain damage or seizures may be predisposing factors; high doses should be avoided in known seizure patients. Several patients have shown sudden flare-ups of psychotic behaviour patterns shortly before death. Autopsy findings have usually revealed acute fulminating pneumonia or pneumonitis, aspiration of gastric contents, or intramyocardial lesions. Although this is not a general feature of fluphenazine, potentiation of central nervous system depressants (opiates, analgesics, antihistamines, barbiturates, alcohol) may occur. The following adverse reactions have also occurred with phenothiazine derivates; systemic lupus erythematosus-like syndrome, hypotension severe enough to cause fatal cardiac arrest, altered electrocardiographic and electroencephalographic tracings, altered cerebrospinal fluid proteins, cerebral oedema, asthma, laryngeal oedema, and angioneurotic oedema; with long-term use - skin pigmentation, and lenticular and corneal opacities. Injections of fluphenazine decanoate are extremely well tolerated, local tissue reactions occurring only rarely.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Symptoms: Restlessness, blurred vision, muscle spasm of face, back and neck. Twitching movement, twisting movements of body. Inability to move eyes, weakness of arms and legs. Difficulty in speaking or swallowing. Loss of balance control. Mask-like face, shuffling walk, stiffness of arms or legs and trembling and shaking of hand and fingers. Hypotension, fainting. Defective colour vision. Lip smacking or puckering, puffing of cheeks, rapid or worm-like movements of tongue, uncontrolled chewing movements and uncontrolled movements of arms and legs.

Treatment of Overdose: Treatment is essentially symptomatic and supportive. The following may be considered: Attempting early gastric lavage. Avoiding induction of vomiting. Administering activated charcoal slurry. Administering saline cathartic. Maintaining respiratory function. Maintaining body temperature. Monitoring cardiovascular function for not less than 5 days. Controlling cardiac arrhythmias with intravenous phenytoin 9 to 11 mg per kg of body weight. Digitalizing for cardiac failure. Administering vasopressor such as norepinephrine or phenylephrine for hypotension. Control convulsions with diazepam followed by phenytoin 15 mg/kg, while monitoring ECG. Administering benztropine or diphenhydramine to manage acute parkinson-like symptoms that may occur.

INCOMPATIBILITIES:

Unknown

STORAGE CONDITIONS:

Store below 30ºC. Protect from light. Avoid freezing.
Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

Pack in 1ml ampoule in 10's and 100's units in a box.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.